



The Gobi Desert

High, harsh, and dry, a desert of climatic extremes, rugged beauty—and very little sand

The Gobi Desert stretches a staggering 500,000 square miles (1.3 million square km) across Asia's remote heartland, making it the continent's second largest desert. Once part of the Mongol Empire, today it straddles northern China and southern Mongolia.

The waterless place

Bordered by Siberia to the north and the Tibetan Plateau to the south, the Gobi lies in the shadow of the Himalayas, which block most clouds laden with life-giving rain. Indeed, the Gobi derives its name from the Mongolian word *gēbi*—the “waterless place.” Seasonal monsoons occasionally moisten its southeast corner, but the desert is mainly arid—average annual rainfall is just 4–6 in (100–150 mm), although snow sweeps into the north



from the Siberian Steppes. Summers are brutally hot, with temperatures soaring to 122–151°F (50–66°C) during the day; in winter, night temperatures can plummet to -36°F (-38°C). Within just 24 hours, temperatures may shift by as much as 60°F (33°C).

Most of the Gobi's terrain is rocky, forming a pavementlike surface (see panel, below), which tempted merchants to cross it as part of the ancient Silk Road. Rugged valleys, chalk or gravel plains, and rocky massifs abound, but sand, for the most part, is thin on the ground. Exceptions include the massive dunes at the base of the Gobi Altai Mountains in southern Mongolia.

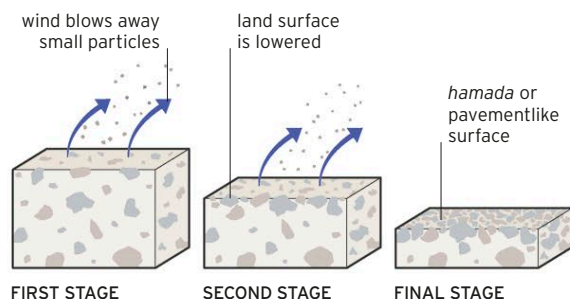
◀ BORN SURVIVOR

Wildlife adapts to the Gobi Desert in remarkable ways. The long-eared hedgehog, which shelters in burrows during the day, can go without food or water for up to 10 weeks.

Only **five percent** of the Gobi Desert is made up of **sand**

HOW A DESERT PAVEMENT FORMS

Water and wind together form hard desert pavements. Over millennia, silt particles on extremely dry ground are either blown away by wind or filtered beneath the surface during the rare occasions when moisture reaches the desert floor. This leaves gravel behind, which fuses into a conglomerate whenever the minerals in it, such as gypsum, absorb enough soil moisture to liquefy, then set. The result is a “pavement.”



△ RED BEDS

The color red dominates the Gobi due to the large amounts of iron oxide present in its rocks. “Red bed” formations are typically made up of sandstone, siltstone, and shale.

▷ SINGING SAND

At Khongoryn Els, Mongolia, “Singing Dunes” at the foot of the Altai Mountains rise as high as 980 ft (300 m). Their name refers to the sound made when wind sweeps the dunes.

